KETAN VASUDEVA

Robotics Engineer

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github.com/natek1234



EDUCATION

Aerospace Engineering - PhD

University of Arizona

January 2026 – Present

- **Thesis:** Multi-modal AI for Autonomous Space Service Robots

Aerospace Engineering - MASc

University of Toronto

September 2023 – August 2025

- Participated in and chaired the Lunar BRICS consortium `https://lunar-bric.arizona.edu/`
- **Thesis:** Lunar Robotic Construction System for Raw Regolith

Engineering Science (Honours) - Robotics Major - BASc

University of Toronto

September 2018 – April 2023

- Minor in **Business** and Dean's List
- **Thesis:** Markov Brain Techniques for Robotic Swarm Control in Space Exploration

EXPERIENCE

Systems Engineer

MDA Space

September 2023 – August 2025 Mississauga, Ontario

- Working in R&D with MDA Space and the University of Toronto to design and develop robotic technologies for lunar construction, gaining expertise in design for harsh environments.
- Conceptualized and performed system-level design for a robotic system to fill and manipulate lunar regolith containment units for the construction of surface structures, performing in-lab testing with regolith simulant, as well as electromechanical design and analysis.

Praxis 3 Teaching Assistant

University of Toronto

January 2023 – March 2025 Toronto, Ontario

- Worked as an Assessment and Studio TA, delivering laboratory sessions focused on software, electronic, mechanical, and system-level design.

Science Software Engineer

GHGSat

May 2021 – May 2022 Montreal, Quebec

- Worked in R&D to deliver industry-level greenhouse gas density maps by implementing scientific software to process raw satellite data.
- Independently developed and maintained a command-line tool to visually analyze and plot satellite retrieval data.

Full Stack Software Engineer

Manulife - Canadian Architecture

May 2020 – August 2020 Waterloo, Ontario

- Developed and deployed a web tool, using **Agile** methodology, to **automate** security threat analysis reports for Security Architecture, **increasing productivity by 30%** by decreasing average report creation time.

PUBLICATIONS

Undergraduate Research

EUCASS Conference: Sustainable Human-Centred Lunar Settlement via ISRU Technologies (**DOI:** 10.13009/EUCASS2022-6167)

Small Satellite Conference:

- FINCH — Accessible Remote Sensing Satellite Missions (**Code:** SSC22-WKVII-04)
- Inspiring the Next Generation — Onboarding & Retention in Undergraduate CubeSat Teams (**Code:** SSC21-P2-37)

Graduate Research

Progress in Aerospace Sciences: Lunar Construction — State-of-the-Art Survey (**DOI:** 10.1016/j.paerosci.2025.101159)

MDPI Aerospace:

- Lunar Robotic Construction System Using Raw Regolith — Design Conceptualization (**DOI:** 10.3390/aerospace12110947)
- Lunar Robotic Construction System Using Raw Regolith — System Engineering (**DOI:** 10.3390/aerospace13060492)

PERSONAL PROJECTS

NASA SpaceApps 2020 Hackathon - Toronto and Canada CSA Winner

- Created an educational sandbox tool to simulate the challenges of communicating with people on Mars using **NASA/CSA** data sets, **React.js**, **Node.js/Express.js**, and **Python**.

The Sound of Space Podcast Host (UTAT)

- Founded, led, and co-hosted a podcast dedicated to explaining the wonders, dangers, and practical applications of space.

University of Toronto Aerospace Team (UTAT)

- Worked in Space Systems for the payload electronics, electrical, systems, and firmware teams to design and build an imaging satellite to measure anthropomorphic greenhouse gas emissions.

UTDL Robodog Software Team Lead

- Led a software development team of 10 for an autonomous robotic dog and manipulator, leveraging **ROS** to tackle challenges in computer vision, path planning, and control systems.

ACHIEVEMENTS

- **Paula Burke Bursary**, **University of Toronto Scholar** and **AIA Canada Scholarship** recipient
- UofT Varsity Ultimate Team **Rookie of the Year**